Sean Yan
64322415
CPSC 320 A2

1 Slow as a logarithm

expanding on $\log_b(n) \in O(n^x)$ with the definition of big O,

$$\exists c \in \mathbb{R}, \exists n_0 \in \mathbb{R}, c > 0$$
$$\forall n \geq n_0 > 0,$$
$$\log_b(n) \leq c \cdot n^x$$

for every $b > 1$ and $x > 0$
let $n_0 = 1$ and $c = 1$

$$\log_b(1) \leq 1 \cdot 1^x$$
$$0 \leq 1 \cdot 1^x$$
$$0 \leq 1$$

so $n_0 = 1$ and $c = 1$ is an valid choosing of c and n_0

and we want, by the definition of little o

$$\lim_{n \to \infty} \left(\frac{\log_b(n)}{n^x} \right) = 0$$

Question 2: I wanna be the very best

a)

this is a combination question.